



Missing & Unbalanced Data

Two Data Quality Problems, Solved Simply

AWS Machine Learning Associate • Part 9

Presented by Prajan Shrestha

What We'll Cover



Why data goes missing



Handling missing data



Choosing the right technique



What is imbalanced data?



Why it's a problem



Fixing it & measuring success

Why Data Goes Missing



Sensor or system failures

Equipment stops recording mid-collection



Optional form fields

People skip questions they don't want to answer



Merging datasets

Some records simply have no matching values



Privacy redactions

Sensitive information deliberately removed

You almost never get a perfectly complete dataset — so you need a plan for the gaps.

Handling Missing Data: Techniques



Deletion

Drop rows or columns with missing values — simplest, but can lose useful data



Mean / Median / Mode

Fill gaps with the average or most common value



Forward / Backward Fill

Carry the last known value forward — common for time series



Predictive Imputation

Use a model (regression, KNN) to estimate the missing value

Choosing the Right Technique

Only a few values missing, no pattern



Deletion is fine

Numeric column, missing at random



Mean / median imputation

Time series data



Forward / backward fill

Missingness relates to other features



Predictive imputation

What is Imbalanced Data?



In a classification problem, one class massively outnumbers another.

Classic example: fraud detection, where roughly 99% of transactions are legitimate and only 1% are fraudulent.

That kind of imbalance is extremely common — and it creates a subtle but serious problem.

Why It's a Problem: The Accuracy Trap



A model that always predicts "not fraud" would be right 99% of the time.

It would look highly accurate — but it would never catch a single actual fraud case, the exact thing you built it to catch.

High accuracy can hide a completely useless model when classes are imbalanced.

Handling Imbalanced Data: Techniques



Oversampling

Add more minority-class examples — duplicate, or generate synthetic ones with SMOTE



Undersampling

Trim down the majority class so the two are more balanced



Class Weighting

Penalize mistakes on the minority class more heavily during training



Collect More Data

Gather more minority-class examples, when feasible

Choosing Better Metrics



Precision

Of your positive predictions, how many were actually correct?



Recall

Of the actual positives, how many did you catch?



F1 Score

A balance between precision and recall



AUC-ROC

How well the model separates classes across every threshold

Accuracy alone is misleading for imbalanced problems — these tell the real story.

Recap

- Missing data: deletion, imputation, or prediction, depending on the case.
- Unbalanced data: resampling & class weighting rebalance the classes.
- Always measure with precision, recall, F1 or AUC-ROC, not just accuracy.

Get both right, and your model is built on a much more honest foundation.

Thanks for watching!